



A PATIENT WITH A CORNEAL DISORDER BEING READIED FOR STEM CELL TREATMENT

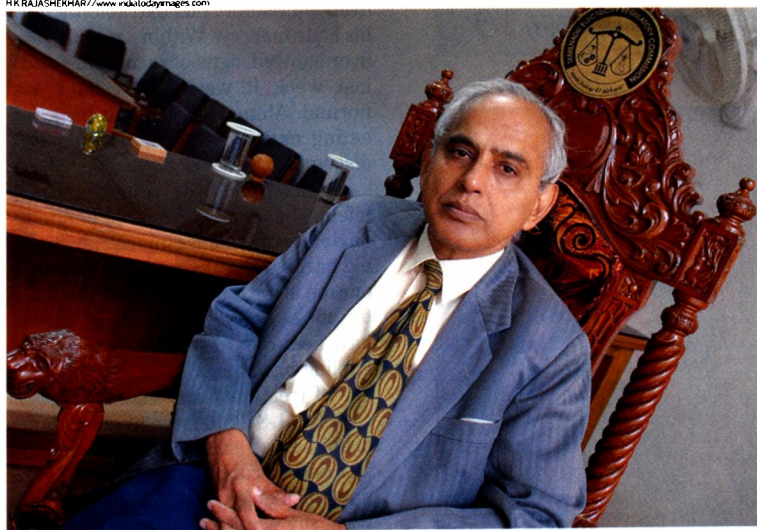
clinical treatment. Though with Shinya Yamanaka, the Japanese stem cell biologist, recently demonstrating that he could derive stem cells of the potency of embryonic ones from skin tissue, it has become a game changer. Ethical debates over its use may soon be a thing of the past if his technique is perfected and it shows that no tumours develop. Karan Goel, organiser of the Stem Cell Global Foundation, estimates that the world market for stem cell therapy is projected to increase from an estimated \$30 billion (Rs 1,44,000 crore) in 2009 to \$96 billion (Rs 4,60,800 crore) by 2015.

At the same time, Indian couples are flocking in droves to cord blood banks to preserve stem cells that could be extracted from the umbilical cord when it is severed after delivery. The steep prices of between Rs 75,000 to Rs 1 lakh to preserve cord blood in cryogenic storage vaults for a period of 21 years hasn't discouraged takers and over a dozen such banks have been set up in all ma-

jor metros. In Gurgaon last week, a couple frantically registered with Cryobanks International for extracting cells of the wife's second pregnancy. Their first child, a boy, was suffering from leukemia and they came to know that cord blood stem cells could be used for treating even the siblings and parents of the baby. Cryobanks CEO Chaitanya Nerikar says that since it was set up three years ago, it has expanded to 75 collection centres and now has over 10,000 samples stored in its vaults. The unit is run almost like a nuclear research facility with tight security and protocols.

The growing popularity of cord blood banks is partly because of the encouraging results coming out from a host of clinical research being done in hospitals across the country. At the Lifeline Multispeciality Hospital in Chennai, Latif Ahmed, a truck driver, is showing all the signs of an astonishing recovery after being diagnosed for cirrhosis of the liver caused possibly by overuse of alcohol. He could barely eat his favourite non-vegetarian food, his blood pressure had dropped and

H.K. RAJASEKHAR//www.indiatodaymagazine.com



HEART DISEASES

S. Kabilan, 62

S. Kabilan, a senior bureaucrat and former chief secretary of Assam, had a bypass surgery at AIIMS, Delhi, four years ago. He was advised stem cell therapy to repair his damaged heart. Within six months of treatment, the amount of blood pumped out of his ventricles with each heartbeat showed considerable improvement. Four years later his heart is in good condition and Kabilan says, "I feel fit as a fiddle."

STATUS IN INDIA

India is among the leaders in the world in the treatment of heart disorders using stem cells. AIIMS itself has treated over 100 cases of both acute heart attacks and cardiomyopathy with some success. Multi-clinical trials are now in five institutes to validate the efficacy of stem cell treatment for heart diseases.



BALRAM BHARGAVA, Professor of Cardiology, AIIMS

“The treatment of cardiac diseases is encouraging but it's not magic. Still, in a field that has few options for patients this is a definite improvement.”